Radiation Shielding & Attenuation Lab

Scientific validation and delivery review / Version 1.0.0

Actual executed checks, traceable NIST source data, independent numerical references and explicit verification limits.

MATERIALS	SOURCE ENERGY ROWS	VITEST PASSED / FAILED	BROWSER CASES NOT EXECUTED
11	1110	169 / 0	30

What was verified

All 169 Vitest cases passed, including 10 independently coded Decimal references. Data checks cover 11 materials, 1,110 source rows, 54 pre/post absorption-edge pairs, exact rows and original-file hashes. The maximum reference absolute difference is 4.44e-16. This measures numerical implementation, not experimental accuracy.

npm ci, TypeScript checking, the production build, offline-data byte reproducibility and execution of the built standalone bundle in jsdom all completed successfully. The archive includes actual machine-readable results, original source snapshots and runnable verification code.

What remains unverified

Thirty Playwright scenarios (15 each for desktop Chromium and mobile Chromium) were enumerated but not run. This environment lacks the browser binary and required browser-control capability. Real browser rendering, responsive layout, canvas PNG encoding and browser-triggered PDF/CSV/PNG downloads remain pending. DOM checks do not substitute for browser tests. No GitHub CI run or deployment is claimed.

Model and rights boundary

Uncollided gamma/X-ray photons only: no buildup, scattered-field transport, secondary radiation, source geometry or dose certification. NIST SRD data retain their own copyright; separate public redistribution permission has not been obtained. Source code is MIT-licensed. This deliverable is prepared for private scientific review.

Ten-calculation reference comparison

Independent Python Decimal calculation at 60 digits. Published rounded coefficients are the input; long displayed regression values do not imply physical precision. Signed difference = application - reference. Arithmetic tolerance: 1e-12 absolute.

Calculation	Equation	Source	Executed test / input	Reference	Application	Difference	Status	Limit of use
01 Mass coefficient	kappa(E)	SRD126 Pb	physics.test.ts reference 01; Pb, 1 MeV; exact SRD126 row	0.07102 cm2/g	0.07102	0	PASS	Source grid; explicit edge side; no extrapolation.
02 Linear coefficient	mu=kappa*rho	SRD126 Pb	physics.test.ts reference 02; Pb, 1 MeV; rho=11.35 g/cm3	0.806077 1/cm	0.806077	-1.11e-16	PASS	Homogeneous sample; density must be specified.
03 Narrow-beam transmission	exp(-mu*x)	SRD126 Pb	physics.test.ts reference 03; Pb, 1 MeV; x=1 cm	0.44660667206 fraction	0.44660667206	0	PASS	Monoenergetic uncollided photons, not dose.
04 Required thickness	-ln(T)/mu	SRD126 Pb	physics.test.ts reference 04; Pb, 1 MeV; target T=0.01	5.7130648635 cm	5.7130648635	0	PASS	Target in (0,1]; no scattering or geometry.
05 Half-value layer	ln(2)/mu	SRD126 Pb	physics.test.ts reference 05; Pb, 1 MeV	0.85990194555 cm	0.85990194555	0	PASS	Monoenergetic standard half-layer, not spectral HVL.
06 Tenth-value layer	ln(10)/mu	SRD126 Pb	physics.test.ts reference 06; Pb, 1 MeV	2.8565324318 cm	2.8565324318	4.44e-16	PASS	Related to HVL through ln(10)/ln(2).
07 Multilayer transmission	exp(-sum(mu_i*x_i))	SRD126 Pb + water	physics.test.ts reference 07; Pb 1 cm + water 10 cm, 1 MeV	0.22018724081 fraction	0.22018724081	0	PASS	Fixed coefficients; not scattered/secondary transport.
08 Mean free path	1/mu	SRD126 Pb	physics.test.ts reference 08; Pb, 1 MeV	1.2405762725 cm	1.2405762725	2.22e-16	PASS	First photon interaction, not a charged-particle range.
09 Partial coefficient sum	coherent+incoherent +photoelectric+pair	XCOM Pb, 1 MeV	physics.test.ts reference 09; Pb, 1 MeV; XCOM coherent=.002991, incoherent=.04993, PE=.01810; pair=0	0.071021 cm2/g	0.071021	0	PASS	One XCOM channel family; rounded residual is reported.
10 Spectrum photon ratio	sum(w*T)/sum(w)	SRD126 water	physics.test.ts reference 10; Water 10 cm; 0.1 and 1 MeV; equal photon weights	0.33721596748 fraction	0.33721596748	0	PASS	Per-line photon weights; no dose or redistribution.

The independent fixtures are included in tests/fixtures/reference.json. The generating Python script imports no application code. HVL, TVL and mean free path are useful descriptors of the same exponential family; ten workflows are not ten independent physical models.

Data, sources and reproduction

Material	Reference density / g cm-3	Composition by mass	SRD126 / XCOM rows
Lead	11.35	Pb=1.0	58 / 74
Aluminum	2.699	Al=1.0	38 / 54
Iron	7.874	Fe=1.0	38 / 54
Copper	8.96	Cu=1.0	41 / 57
Tungsten	19.3	W=1.0	59 / 75
Water, Liquid	1.0	H=0.111898; O=0.888102	36 / 52
Concrete, Ordinary	2.3	H=0.0221; C=0.002484; O=0.57493; Na=0.015208; Mg=0.001266; Al=0.019953; Si=0.304627; K=0.010045; Ca=0.042951; Fe=0.006435	46 / 69
Polyethylene	0.93	H=0.143716; C=0.856284	36 / 52
Air, Dry (near sea level)	0.001205	C=0.000124; N=0.755268; O=0.231781; Ar=0.012827	37 / 54
Carbon, Graphite	1.7	C=1.0	36 / 52
Silicon	2.33	Si=1.0	38 / 54

SRD126: https://physics.nist.gov/PhysRefData/XrayMassCoef/cover.html

Lead: https://physics.nist.gov/PhysRefData/XrayMassCoef/ElemTab/z82.html

Water: https://physics.nist.gov/PhysRefData/XrayMassCoef/ComTab/water.html

XCOM: https://physics.nist.gov/PhysRefData/Xcom/html/xcom1.html

Methods/thresholds: https://physics.nist.gov/PhysRefData/Xcom/Text/chap3.html

Reference density/composition: NIST XrayMassCoef/tab1.html and tab2.html

Rights: https://www.nist.gov/open/license; see DATA_LICENSE.md

All source tables were retrieved on 2026-10-14. SRD126 data content is identified as July 2004; the online XCOM response has no per-response release identifier. It is retained as a hashed snapshot, not relabeled as downloadable XCOM 3.1. The 1,110 rows include both source families and both limits at absorption edges.

Uncertainty and execution guide

Uncertainty separation

No per-value NIST standard uncertainty is fabricated. Optional primary density/thickness standard uncertainties yield a first-order independent input component: u(tau)^2=(kappa*x*u_rho)^2+(kappa*rho*u_x)^2, and u(T)/T approximately equals u(tau). It is not a total uncertainty and does not include data, interpolation, geometry or model omissions. No stack/spectral uncertainty is inferred.

Interpolation

Exact rows return published values. Between positive values the local method is segmented log-log; both edge limits are preserved, default edge side is explicitly above, and no energy extrapolation occurs. Pair channels use their own threshold and a locally documented onset. The method is not NIST XCOM's internal shell/spline implementation.

Numerical behavior

The engine uses double precision, expm1 for small losses and log-sum-exp for spectra. Very small transmissions retain logarithms and scientific text; numeric zero from underflow is not treated as physical zero. Most interface/PDF outputs show four significant digits; regression references deliberately show more digits only for arithmetic comparison.

Open and reproduce

Double-click START-HERE.html for the offline app. For source checks: npm ci; npm run validate:data; npm test; npm run typecheck; npm run build; npm run smoke:standalone; npm run validate:scientific. Reconstruct the database with python3 scripts/import_nist.py --offline. Actual browser checks require Chromium: npx playwright install --with-deps chromium; npm run test:e2e.

Publication

No repository or public site was created. Default GitHub Pages base: /RadiationShielding-Lab/. CI and a manual approval-gated Pages workflow are included. Review data redistribution rights before public release. The archive is a reviewable local deliverable, not a radiation-safety approval.